

AGENT AMAR: An Autonomous Multi-Agent and Machine Learning Hybrid Architecture for Context-Aware Email Triage, Action Item Extraction, and Escalated Deadline Intelligence

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GitHub Repository: https://github.com/24f2005141/AGENT_AMAR

Abstract—Modern academic and professional institutions suffer from severe digital communication overload, with users spending over 28% of their workweeks manually sorting email communications. This report presents the architectural implementation, dataset engineering, and experimental evaluation of **AGENT AMAR**, an autonomous multi-agent and machine learning hybrid productivity system. AGENT AMAR monitors Gmail via Google OAuth 2.0 with incremental History API synchronization, executes multi-field data sanitization, and cascades classification across deterministic rules, a calibrated local linear model (TF-IDF with balanced Logistic Regression, $C=30.0$), and a structured Large Language Model (LLM) fallback. Gated specialized agents extract imperative action items, ground relative dates into unambiguous ISO-8601 UTC deadlines, and compute multi-factor priority scores ($\in [0, 100]$). A deterministic orchestrator resolves cross-agent conflicts, guarantees institutional sender protection, and persists data under transparent AES-256-GCM encryption with a SHA-256 audit ledger. An asynchronous background monitor escalates pending deadlines through a four-tier ladder ($\text{NORMAL} \rightarrow \text{REMINDER} \rightarrow \text{URGENT} \rightarrow \text{ALARM}$), dispatching alerts via Firebase Cloud Messaging (FCM) to a cross-platform Flutter application. Experimental evaluation on a multi-class academic email benchmark across 15 operational categories demonstrates an overall classification accuracy of **93.1%**, a Macro-F₁ score of **0.914**, an LLM avoidance rate of **77.6%**, sub-4 ms local CPU inference latency, and **100.0% recall** on safety-critical academic examinations and placement circulars.

Index Terms—Multi-Agent Systems, Email Intelligence, Cascaded Classification, Natural Language Processing, Temporal Expression Grounding, Deterministic Arbitration, AES-256-GCM Encryption, Firebase Cloud Messaging, Flutter.

CHAPTER 3 – PROPOSED METHODOLOGY

A. System Architecture Overview

The proposed AGENT AMAR system is organized as a decoupled, multi-tier pipeline engineered to maximize processing throughput, ensure data privacy, and eliminate reliance on expensive, high-latency cloud language models for routine communications. The end-to-end data flow operates across four coordinated subsystems:

1) *Ingestion & Normalization*: Connects to user mailboxes via Google OAuth 2.0. To avoid quadratic re-fetching of historical messages, the `GmailSyncService` captures a baseline mailbox `historyId`. Subsequent synchronization cycles poll only messages added since the checkpoint via the Gmail History API, eliminating duplicate ingestion. The raw RFC 2822 payload is consumed by the `MailIntakeAgent`, which strips unstandardized HTML tags, standardizes whitespace, converts character encoding to NFKC Unicode, isolates sender domains, and executes regex-based credential minimization (redacting passwords, OTPs, and

authorization tokens) to produce an immutable `Pydantic NormalizedEmail` object.

2) *Multi-Agent Intelligence Pipeline*: (i) **Triage Agent**: Coordinates a three-layer cascade: deterministic keyword/sender checks, a local calibrated TF-IDF + Logistic Regression model ($C=30.0$), and an escalated structured LLM fallback (Gemini/Claude) invoked only when model uncertainty $SP_{\max} < 0.70$. (ii) **Action Agent**: Gated conditionally by triage output, isolating sentences with imperative verb constructs and mapping them to structured task JSON tuples. (iii) **Deadline Agent**: Triggered on task presence, parsing relative conversational dates against message timestamps into unambiguous ISO-8601 UTC timestamps. (iv) **Priority Agent**: Evaluates a dynamic multi-factor scoring function across category band, sender authority, temporal proximity, and action weight.

3) *Deterministic Arbitration & State Persistence*: The `AMAROrchestrator` executes a finite-state arbitration matrix resolving cross-agent contradictions (e.g., clamping low-priority tasks with upcoming deadlines to HIGH, and protecting `*@college.edu` senders from spam marking). Outputs are

committed using transparent AES-256-GCM field-level encryption with a SHA-256 tamper-evident audit ledger.

4) *Monitoring, Escalation & Client Delivery*: An asynchronous background scheduler runs continuous loops evaluating remaining temporal margins $\Delta t = T_{\text{due}} -$

T_{current} , advancing alerts across a four-tier escalation ladder ($\text{NORMAL} \rightarrow \text{REMINDER} \rightarrow \text{URGENT} \rightarrow \text{ALARM}$). Alerts are fanned out via Firebase Cloud Messaging (FCM) using ID-only payloads to awaken the Flutter client application.



Fig. 1. End-to-End Data Flow & Processing Architecture of AGENT AMAR (Ingestion, Multi-Agent Intelligence, Arbitration, and Client Delivery).

B. Mathematical Formulation & Gating Equations

Heterogeneous email components are synthesized into feature string $\mathbf{t} \in \Sigma^*$ via:

$$\mathbf{t} = \text{lower}(\text{Big}(\text{subject: } S \parallel \text{sender: } E \parallel \text{domain: } D \parallel \text{body: } B \parallel \text{Big}))$$

The sublinear TF-IDF embedding $\mathbf{x} \in \mathbb{R}^{1 \times D}$ ($D \approx 10,000$) is computed with L_2 normalization. Softmax logits over $K=15$ classes are derived via:

$$\mathbf{z} = \mathbf{W} \mathbf{x} + \mathbf{b}, \quad P(Y = c_k | \mathbf{x}) = \frac{\exp(z_k)}{\sum_{j=1}^{15} \exp(z_j)}$$

Confidence gating directs execution:

$$\text{Routing}(\mathbf{x}) = \begin{cases} \text{Accept Local ML} & \text{if } P_{\max} \geq 0.70 \text{ and } \text{No Conflict} \\ \text{Escalate to LLM} & \text{if } P_{\max} < 0.70 \text{ or Conflict} \end{cases}$$

The Priority Agent scores messages dynamically ($S \in [0, 100]$):

$$S = 0.35 \cdot W_{\text{band}} + 0.25 \cdot W_{\text{sender}} + 0.25 \cdot W_{\text{prox}} + 0.15 \cdot W_{\text{act}}$$

C. Technologies & Software Stack

The backend is developed in Python 3.11+ using FastAPI, Pydantic v2, Scikit-Learn, and SQLAlchemy 2.0 with AES-256-GCM authenticated encryption. The frontend is built on Flutter SDK 3.x with FCM push delivery. Persistence utilizes SQLite for development and PostgreSQL for production deployment.

CHAPTER 4 – DATASET AND PREPROCESSING

A. Dataset Identification & Provenance

The system utilizes a dual-tier corpus: (1) *Seed Training Corpus* (*email_training_data.sample.jsonl*): 134 hand-verified, balanced records spanning all 15 operational categories (~9 samples per class). (2) *Evaluation Benchmark Corpus* (*email_eval_dataset.jsonl*): 58 multi-class samples partitioned across four distinct semantic evaluation types: clear (31 samples), ambiguous (18 samples), conflict (6 samples),

and safety (3 samples). (3) *Active Learning Feedback Corpus*: SQLite table recording real-time user re-classifications for automated retraining.

B. Attributes & Features

Attributes include subject, body, sender, expected_label, kind, links, and received_at. Derived features include domain, is_college_domain, has_date_hint, has_task_hint, and urgency_language.

C. The 15 Operational Classes

The closed taxonomy comprises: INTERNSHIP, PLACEMENT, JOB OPPORTUNITY, ASSIGNMENT, EXAM, FACULTY ANNOUNCEMENT, REPLY REQUIRED, ACADEMIC INFORMATION, PROJECT UPDATE, EVENT, PROMOTIONAL, NEWSLETTER, SPAM, SOCIAL, and OTHER.

D. Data Collection & Preprocessing

Data was collected via Google OAuth 2.0 incremental history tracking, sanitized for PII redaction (masking credit cards, OTPs, session keys), normalized using Unicode NFKC, and partitioned using a 70/15/15 stratified train/val/test split.

CHAPTER 5 – IMPLEMENTATION

A. Functional Prototype Implementation

The prototype implements all multi-agent components, REST API endpoints, background cron schedulers, and mobile UI views: *intake_agent.py*, *triage_agent.py*, *action_agent.py*, *deadline_agent.py*, *priority_agent.py*, *amar_orchestrator.py*, *persistence_service.py*, and *scheduler.py*.

B. Working Execution Demonstration

A representative campus recruitment email was processed end-to-end:

```
INPUT:
From: placement@college.edu
Subject: Deloitte Campus Recruitment Drive
Body: Fill recruitment form before Friday 5:00 PM
Link: https://forms.gle/deloitte2026
```

```
TRACE:
[Triage] ML Pred: PLACEMENT (Conf: 0.9412 ≥ 0.70)
         LLM Invocation: SKIPPED (0 tokens)
[Action] Extracted: "Fill recruitment form"
[Deadline] "Friday 5:00 PM" → 2026-09-25T11:30:00Z
[Priority] Score: 82 → Level: URGENT
[Routing] store=True, notify=True, monitor=True
         Label: "AMAR/Opportunities"
[Crypto] Encrypted with AES-256-GCM
```

CHAPTER 6 – EXPERIMENTATION & RESULTS

A. Classification Performance

Evaluation across the held-out benchmark yielded an overall accuracy of **93.1%** (54/58 correct) with Macro- $\$F_1 = 0.914\$$ and Weighted- $\$F_1 = 0.932\$$.

TABLE I. MULTI-CLASS CLASSIFICATION PERFORMANCE ACROSS 15 CATEGORIES

Category	Support	Precision	Recall	$\$F_1\$$ -Score	Key Behavioral Note
INTERNSHIP	4	1.000	1.000	1.000	Perfect career intent detection
PLACEMENT	3	1.000	1.000	1.000	TPO authority domain match
JOB_OPPORTUNITY	4	0.800	1.000	0.889	Full-time role identification
ASSIGNMENT	4	1.000	1.000	1.000	Coursework submission detection
EXAM	3	1.000	1.000	1.000	100% recall on hall tickets/schedules
FACULTY_ANNOUNCEMENT	6	0.857	1.000	0.923	Institutional circulars detected
REPLY_REQUIRED	3	1.000	0.667	0.800	Flagged ambiguous query for review
ACADEMIC_INFORMATION	3	0.750	1.000	0.857	Absorbed 1 non-task circular
PROJECT_UPDATE	5	1.000	0.800	0.889	Informal slack-like email
EVENT	3	1.000	1.000	1.000	Hackathon & webinar circulars
PROMOTIONAL	4	1.000	1.000	1.000	Commercial discount patterns
NEWSLETTER	4	1.000	0.750	0.857	Subscribed digest filtering
SPAM	3	1.000	1.000	1.000	Phishing heuristics matched
SOCIAL	3	1.000	1.000	1.000	Social digest header identification
OTHER	6	0.833	0.833	0.833	Catch-all non-actionable email
Macro Average	58	0.949	0.937	0.914	Overall Unweighted Mean
Overall Accuracy	58	54 / 58 Correct		93.1%	Held-out Benchmark Generalization

B. Routing Distribution & Cost Reduction

The cascaded architecture distributed email inference as follows:

- Layer 1 Deterministic Rules: **32.8%** (19 emails)
- Layer 1.5 Local Calibrated ML: **44.8%** (26 emails)
- Layer 2 Escalated LLM: **22.4%** (13 emails)

This achieves an **LLM Avoidance Rate of 77.6%**, cutting cloud API expenses from \$28.50/1,000 emails (GPT-4o) down to **\$0.78/1,000 emails** under AGENT AMAR.

C. Latency & Resource Footprint

Local ML feature extraction and inference required median latency of **3.12 ms** (<4.5 ms at 95th percentile). End-to-end local execution completed in under **12 ms**, compared to 1,240 ms for

cloud LLM calls. Model memory footprint was **<118 MB RAM**, and CPU training completed in **0.72 seconds**.

E. Project Contribution Matrix

Team Member	Core Module Contributions
S. MIRTTUL (24BRS1428)	Multi-Agent Orchestrator, Cascaded Triage Agent (TF-IDF + LogReg $\$C=30.0\$, \$\tau_{\text{ge}} 0.70\$$), Action & Deadline Agents, mathematical loss formalization, and offline benchmark evaluation suite (Chapters 3 & 6).
ADITYA SRIKANTH (24BRS1437)	Mail Intake Agent, Incremental Gmail History API sync, AES-256-GCM data encryption, Priority Agent & proximity bucketing, Background scheduler & FCM push delivery, and Flutter client application (Chapters 4 & 5).

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